



Contents lists available at ScienceDirect

Journal of Rail Transport Planning & Management

journal homepage: www.elsevier.com/locate/jrtpm

The impact of railway incidents on train delays: A case of the Swedish Railway Network

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ABSTRACT

A foreseeable challenge with a substantial increase in railway mode share will be how to uphold punctuality. Higher volumes of train traffic will result in timetables that are more sensitive to disruptions; whose severity and frequency is also expected to increase in light of greater asset utilization and climate change. This calls for a definitive understanding of the relationship between incidents and train delays as a prerequisite to developing robust timetables and disruption management strategies. In this paper we propose a novel framework for quantifying the impact of railway incidents on train delays. Using a case of the Swedish Railway Network, we compare the impact of different incidents on train delays. The impact of delay is defined as a factor of the incident rate, exposure rate, delay rate and historical average delay minutes per incident. A logistic model that estimates the probability of delay for any train, in the event of a failure, is also developed. Snow on track was established as most critical, resulting in the highest normalized delay minutes per train and the largest increase in the odds of delay for individual trains. The proposed framework & approach can be applied to other networks.

1. Introduction

The targeted growth in mode share of railways (Directive, 2012/34/EU) as a response to the global climate change crisis, presents several potential challenges to the railway industry. One such challenge will be how to uphold and/or achieve high levels of operational reliability, in particular punctuality. As train headways are decreased to accommodate more trains (closer running), the timetable will become more sensitive to incidents on the network. Especially secondary (knock-on) delays, and primary delays will increase both in magnitude and reach. Secondary delays are caused by delayed preceding trains, while primary delays are those introduced into the timetable by unexpected/random events occurring on the network i.e., incidents (Sipilä, 2015). The presumed increase in primary delays is associated with the incumbent and projected climate change trends where extreme weather events which are either responsible for or precipitate the bulk of exogenous incidents continue to increase in frequency and severity (IPCC, 2023; Xia et al., 2013). Evidently, an inharmonious situation in which an increase in incidents - a source of delays/disruptions is met with the shrinkage of the timetable's ability to 'absorb' delays is expected. Highly efficient disruption management, robust timetables, and good operations planning will, therefore, be necessary to contain the situation. This cannot be achieved without a good understanding and quantitative definition of the relationship between railway incidents and train delays.

So far, research in this area, has had a predominant focus on resilience i.e., the railway system's ability to continue to provide a service in the event of a disruption-a definition that is adopted from the European Committee for Standardization (CEN, 2021). Bešinović (2020) cites four main approaches that are used to quantify resilience i.e., topological methods based on network theory, simulation, optimization and the data driven approach. He cites the data driven approach as the least demanding given that there is no

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need to explicitly model the system. In this approach, resilience is measured and quantified as the deviation from the standard level of service during disruptions - derived from historical operations data, (Fabella and Szymczak, 2023). However, most of the related studies in railways quantify this deviation in terms of traffic intensity, and in so doing account only for cancelled trains - disregarding punctuality effects. Moreover, past studies have mainly focused on a few extreme events, typically natural hazards, paying little attention to the more commonly occurring incidents from technical failures. For instance, Fabella and Szymczak (2021,2023) estimated the changes in daily traffic volumes on the Germany railway network during floods, mass movements, slope fires, and tree falls. Knoester et al. (2024) overcomes both limitations by proposing a performance index that combines both punctuality and cancellations, and also focuses on the more regularly occurring incidents i.e., vehicle failures, signal failures, switch failures, overhead line failures and collisions. However, in this study the proportion between cancelled and punctual trains is subjective suggesting that the gap in understanding the evolution of delays to trains during incidents still exists. Additionally, on the whole, the resilience metric aggregates the impact of an incident over the traffic using the network at the time, and the effect on an individual train is masked. While this may be sufficient for disruption management, it may be hard to interpret and use in other applications such as delay prediction conditional upon a given incident occurrence. The metric used also assumes that all delays occurring during a given incident are solely attributable to the incident, which is not always the case. This is no different with punctuality evaluation studies that measure impact in terms of trains delayed and/or additional delay minutes (Xu et al., 2016; Ferranti et al., 2016; Xia et al., 2013).

Moreover, most of the above and other studies investigating incidents' impact on punctuality use delay attribution data (Greenham et al., 2020; Kristoffersson, 2019; Nyström and Kumar, 2003; Wen et al., 2017; Wen et al., 2020). The limitation of the use of delay attribution data is that it only accounts for trains delayed beyond the respective network's threshold for the definition of delayed trains. For instance, in Sweden, delay attribution is only done for trains delayed by 3 min or more (Trafikverket, 2022). Small/sub-threshold delays which constitute a reasonable proportion of delays, especially secondary delays, remain unaccounted for. In a study on factors influencing the punctuality of the Swedish Railway Network in the year 2015, Palmqvist et al. (2020) reported that 'small' delays amounted to 55% of delays. Additionally, according to Nyström (2008), there is low consistency in the delay attribution process. Different technicians are likely to attribute the same delay to different factors. This makes the data and any relationships thereof unreliable.

1.1. Aim of the study

Therefore, this study proposes a framework for quantifying the impact of railway incidents on train delays that takes into account all delays including small/sub-threshold delays, and one that isn't reliant on delay attribution data. The study seeks to evaluate the impact on individual trains in a way that accounts for other factors influencing punctuality. In demonstration of its application, the study uses the proposed framework to estimate and compare the delay impact of different incidents on the Swedish Railway Network. It addresses the following specific research questions.

1. For a given number of trains on the network, what is the expected amount of delay minutes attributable to different incidents? Herein lie further questions including:
 - a) What is the frequency of different events, herein referred to as the incident rate?
 - b) What is the rate of exposure of trains to these incidents? Exposure being the proportion of trains that travel over a section with an¹ active incident.
 - c) What percentage of the exposed trains are delayed for each incident?
 - d) What is the average size of delay per incident?
2. What is the risk of delay for any train travelling on the network when a given incident occurs?

1.2. Structure of paper

The rest of the paper is divided into six main sections. Section 2 presents findings from a literature review conducted with the goal of understanding the definition, measurement and state of the art in train punctuality/train delay research with specific focus on the factors that influence train punctuality. Section 3 presents the case-the Swedish Railway Network; highlighting system characteristics that may influence failure patterns. Section 4 describes the data, its sources, and synthesis. Section 5 describes the methodology for analysis while Section 6 presents and discusses the results and finally, the paper is concluded in Section 7.

2. Literature review

2.1. Punctuality

Punctuality evaluated at the network level is often used as a performance indicator for railway operations. Reference is made to the Public Performance Measure in the UK and the Combined Performance Measure in Sweden. Both indicators combine reliability and punctuality into a single measure evaluated as the proportion of trains that arrive at their final station on time. (NetworkRail, 2023;

¹ Active refers to an incident that is yet to be fixed.

Olsson and Haugland, 2004; Trafikanalys, 2023).

Other countries adopt a similar approach albeit differing in the choice of station for evaluation, the train timestamp for evaluation i.e., departure or arrival time, and/or the size of threshold for definition of on-time performance. In Europe, as elaborated in Schittenhelm, (2011), Denmark, the Netherlands and Germany define some of the most stringent thresholds, while Italy has the most slack definition of on-time performance. A threshold of within 2 min arrival measured at selected stations is defined for sub-urban trains (S-Bahn) in Germany. Copenhagen defines a slightly higher threshold of within 2.5 min arrival and departure at all stations for sub-urban trains in Copenhagen. Both countries define a threshold of 5 min for long distance trains as measured in the suburban trains. Contrary, in the Netherlands the same threshold of within 3 min of arrival at selected larger stations is defined for both long distance and regional passenger trains. This approach is slightly similar to Italy which defines the same threshold i.e., 15 min for both regional and passenger trains; albeit measured for arrival at the end of the line. Sweden also defines a delay threshold of within 5:59 min arrival at the final station for all types of passenger trains i.e., commuter, regional and long-distance trains (Trafikanalys, 2023). Other European countries such as Austria, Lithuania, Poland, the United Kingdom also define a threshold of 5 min for passenger trains (Grechi and Maggi, 2018). Generally, much larger thresholds are defined for freight trains, take an example of 10 min in Denmark, and 30 min in Germany.

Arguably, while network punctuality is useful for providing a general overview of reliability performance, a lot of information is obscured within this single measure. Trains may be delayed at several points during their journeys and still manage to recover from the delays before the designated point of measurement. Evaluating delays at every point in the network therefore seems like a better approach, especially where the goal is to improve punctuality and/or arrest delay-causing factors. Moreover, it allows for more transparent and easily comparable values across different networks.

2.2. Train delays

2.2.1. Primary vs secondary delays

Train delays refer to the deviation between the train's scheduled and actual arrival or departure time from a station. Depending on the cause of delay, a delay may be classified as primary or secondary. Primary delays are train specific and are caused by factors external to the timetable, such as a vehicle failure, while secondary/knock-on delays are caused by train interactions and are propagations of delays from other trains (Röbber et al., 2021; Sipilä, 2015). Palmqvist et al. (2023) estimated that approximately 36% of delays at the network level in the Skåne-region of Sweden are primary while 64% are secondary. This can be interpreted as, on the Swedish Railway Network, a primary delay results in approximately double the amount of secondary delays. This highlights the importance of limiting primary delays in managing the punctuality of a network.

2.2.2. Use of AI to quantify delay impact

It thus follows that the bulk of studies seeking to quantify the impact of incidents on train delays focus on primary delays, after which secondary delays may be inferred. Röbber et al. (2021) used explainable AI to this end, however his model only adopted temporal factors (day of the week and the season) and weather as the only factors driving primary delays. No consideration was given to other system factors. Malavasi and Ricci (2001) also employed big data techniques to extract primary delays, considering only infrastructure failures. The consequence of infrastructure failure events was simulated in terms of number of delayed trains and delay per delayed train on the railway system using a Bayesian neural network model. However, the study made no distinction between different failure types making it difficult to distinguish the impact of different failures. Marković et al. (2015) applied Support Vector Regression analysis and Artificial Neural Network models to the analysis of the impact of railway system characteristics (infrastructure design and condition) on train delays. However, interviews with experts were used to estimate causality relations which arguably introduces subjectivity in the results. In general, the use of big data techniques for this endeavor is still limited by the need for large volumes of data, and the fact that results are not always interpretable.

2.2.3. Use of statistical methods to quantify delay impact

Hence, the bulk of existing studies (both industry and research) instead employ statistical methods on delay attribution data to estimate delay impact. Kristoffersson (2019) estimated the impact of each delay cause on the Swedish railway network as the proportion of total delay minutes attributed to it. The study categorized sources of primary delays into two groups i.e., errors in the functionality of the railway system (group 1) and inefficiencies in the planning and/or control of the railway system (group 2). The former constituted infrastructure failures, vehicle errors, and unauthorized access to track; while the latter included traffic and resource planning, operational traffic management, track work, departure time from nodes and the airport service. Most studies adopt more or less similar classifications with a tendency to disaggregate the infrastructure class further. Nyström and Kumar (2003) adopted a similar approach for Sweden's Iron ore-line: a single line track that carries mixed traffic but disaggregated the infrastructure failure class further into overhead wire (catenary), track, turnouts, signal boxes and section blocks, positioning system, feeder line, shunting equipment, centralised traffic control, level crossing and embankment failures. The study also included secondary delays not attributed to primary delays as a separate class.

Yang et al. (2019) and Xu et al. (2016) also used delay attribution data to study and model delays on specific HSR lines in China. Similar classifications were adopted however unlike in the aforementioned statistical studies, these studies report impact in terms of delayed trains estimated using a zero-truncated negative binomial modal. Yang et al. (2019) further models distributions of delay time for each delay cause. The use of different classifications and different definition of impact (delayed trains, delay minutes etc) makes results across networks incomparable. This study hence bases classification on Sweden's delay cause codes (Trafikverket, 2022) and

attempts to define impact in a way that takes into account number of trains and delay minutes.

2.3. Other factors influencing punctuality

Additionally, besides the use of delay attribution data and the associated limitations as discussed in Section I, most of the above studies explicitly or implicitly assume that all delays to a train are linked to a single event/disturbance. No consideration is made for other (random) factors that may introduce delays in a train's journey - that aren't particularly linked to a system failure.

Several studies investigate some of these factors. [Harris and Godward \(1992\)](#) established that factors such as the distance covered from the departure station and the number of carriages (as the train length) also impacted train punctuality. The latter findings align with [Olsson and Haugland \(2004\)](#)'s work that established that the number of travelers and the load factor of the train also impact punctuality. This same study also established departure punctuality as one of the most influential factors for punctuality. Notably, other factors of influence such as capacity utilization and speed restrictions were deduced, however were seen to only have an influence above a certain threshold.

[Palmqvist et al. \(2020\)](#) also pointed to other factors of influence such as temperature extremes, the size of run time margins in the timetable, and the number and distance between infrastructure elements. According to this study punctuality drops exponentially as the temperature drops below 0 °C. The study also determined that a timetable that includes 25% minimum run time margins or 12 s/km placed towards the end of the journey results in better punctuality. An increase both in the numbers of a particular infrastructure type and the distance between the elements was linked to reduced punctuality by the same study. Arguably, the reported impact of infrastructure could also be linked to the probability of failure. More infrastructure components increase the chances of a train being exposed to the consequences of a component failing e.g., temporary speed restrictions among others. While reduced inter-distances between components i.e., areas with denser infrastructure provide more redundancies. This makes the aspect of 'exposure' (as explained later) crucial in evaluating the impact of incidents on train delays.

As far as the authors know, only one recent study ([Joborn and Ranjbar, 2022](#)) accounts for trains delayed by factors other than incidences/disturbances i.e., trains delayed even when there are no disturbances to the network. The study refers to these as 'minor delays' and established that for the case study, they were responsible for 39% of unpunctual trains (delayed beyond the threshold of 3 min) thus pointing to their significance. Hence using a different approach, this study models the impact of incidents on a train, taking into consideration other factors that may factor delays besides the incidents.

3. Case study

3.1. Track type

The Swedish railway network, as of 2021, consisted of approximately 10,912 km. Approximately 81% is single track while the remaining 19% is either double or multiple track ([Fig. 1](#)).

3.2. Electrification and signaling

While only 69% of the single-track lines are electrified, all the double track is electrified amounting to ca 75%² electrification of the Swedish railway network. Trains receive power supply through a catenary producing 15,000V, 16 2/3 Hz. ([Fig. 2](#)). Heavily electrified railway systems are characterized by a large volume of equipment/components which, as a disadvantage, all constitute potential points of failure in the system. This is also the case with ATC - the predominant standard of ATP (Automatic Train Protection) in Sweden. Well over 90% of the network is installed with ATC system 2 ([Fig. 2](#)).

4. Data

In the proposed framework, two sets of historical operation data are required i.e., a record of all incidents/failure events across the network and train movement data for the same period. For the Swedish railway network, historical operation data for the period between 2017 and 2021 was used. This was obtained from the Swedish Transport Administration (Trafikverket).

4.1. Incident data

Incident reports, as stored in Sweden's failure database constitute a brief description of the incident (including a delay code), the location of the incident, the probable cause, actions taken to remedy the incident, and timestamps related to the rectification process i.e., when the incident was reported, when a team arrived at the incident site and when the incident/failure was fixed. Most relevant to this framework are the times the incident was reported, and when it was fixed, plus the location, the incident type, and the respective delay code.

In total, 290 628 incidents were recorded over the study period. These were classified into 13 main categories, in accordance with

² Rail | European Alternative Fuels Observatory (europa.eu).



Fig. 1. Sweden's railway network: Track types (Top) (Source: <https://jnbkarta.trafikverket.se/?year=2022><https://jnbkarta.trafikverket.se/?year=2023>).

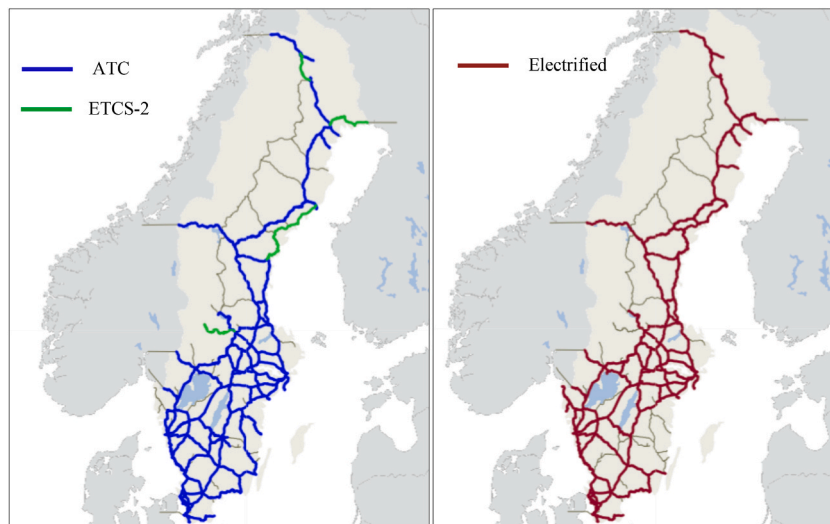


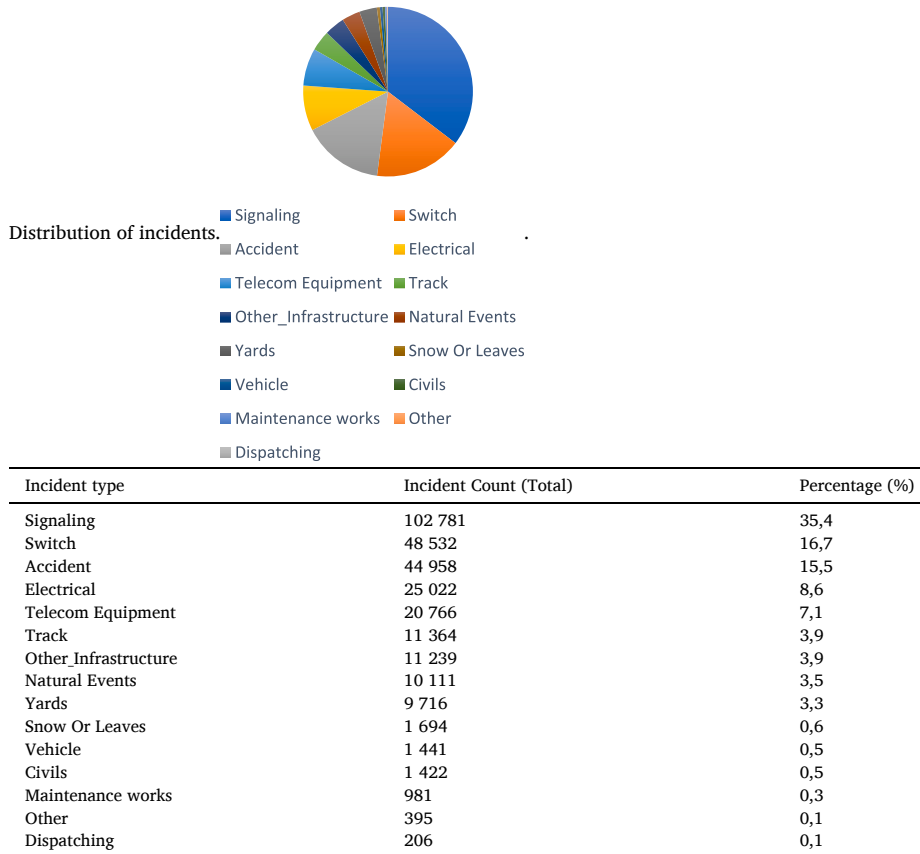
Fig. 2. Sweden's Signaling system (Left), Traction type (Right) (Source: <https://jnbkarta.trafikverket.se/?year=2022> <https://jnbkarta.trafikverket.se/?year=2023>).

the standard delay codes used by the Swedish Transport Administration (Trafikverket, 2022). Table 1 shows the distribution of these incidents across the different classes. Accordingly, signal failures had the highest frequency while dispatching errors had the lowest.

4.2. Train run data

The second set of data: train movement data included the scheduled and actual train departure and arrival timestamps for all trains

Table 1



within the study period; recorded on a minute-by-minute basis at all stations within the network. Considering each station (A-A) and stretch (A-B), for every day of the study period, the total number of trains passing, and the total number of trains delayed was extracted resulting in 2 582 297 unique records.

5. Methodology

5.1. Data synthesis

Combining the failure report data and the train run data, a record of the condition of the infrastructure (i.e., the absence or presence of an active³ incident, classified by type) for everyday of the study period – which by extension is the condition of infrastructure during respective train runs, was extracted. Fig. 3 shows the logic for combining the datasets.

Important to note is that failures/incidents in the database are reported as occurring either at a station or along a stretch. Precise information on the geographic location of these incidents such as geographic coordinates or the track numbers and chainages were missing. Consequently, dwell delays at a given station are matched to failures at the same station (occurring within the same period), and similarly run delays along a given stretch matched to failures along the same stretch. In essence, this assumes that for double or multiple track sections and at stations, a train may still be affected by a failure on another track, besides the one it is running over, provided it is within the same section/station. The matching also assumes that the impact of a failure beyond the section or station within which it occurs is negligible. While the former may result in an overestimation of impact, the latter may result in an underestimation of impact. This points to the need for better and more detailed incident reporting data for a more accurate application of this framework.

5.2. Incident rate

The daily rates of occurrence for the different incidents were computed as a quotient of the total number of failure records and the

³ Active incident refers to one that has not yet been fixed.

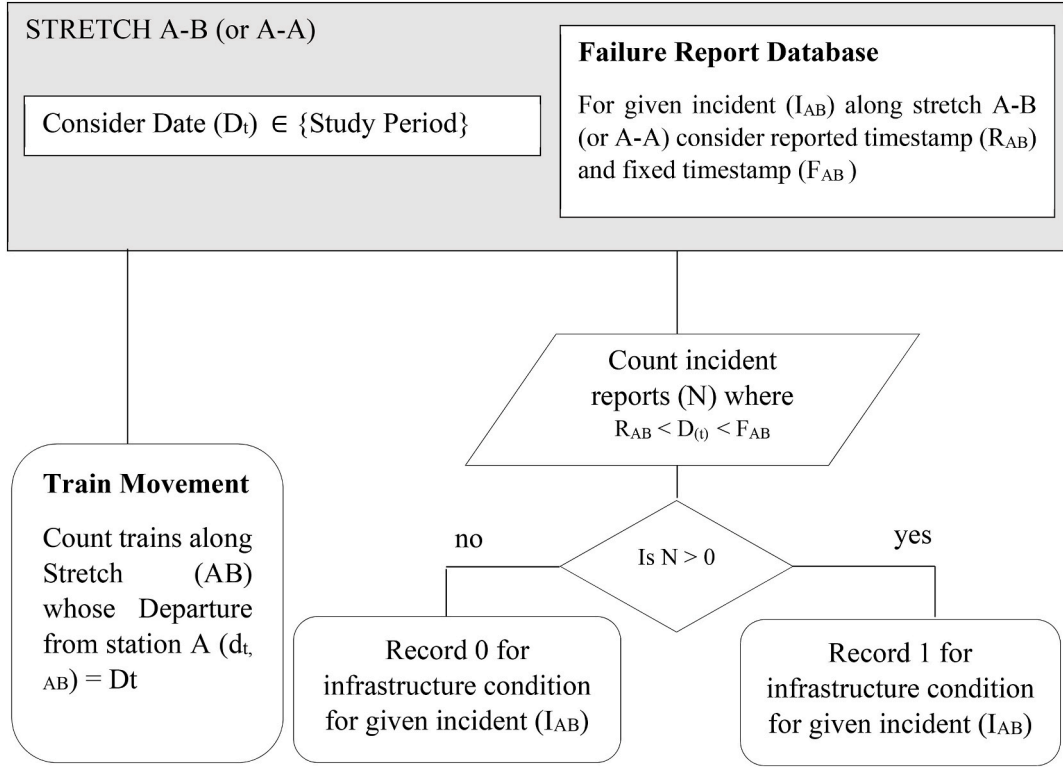


Fig. 3. Logic for deriving the incident status during a train run.

total of days within the study period i.e., 1807 days.

5.3. Train exposure and delay impact

The rate of exposure was estimated as a ratio of the number of trains that run over a given section/station when there is an active incident to the total number of trains that travel over the same section for the period of study (Equation (1)).

$$Exposure_{(AB)} = \frac{\sum_{i=1}^n T_{AB,I=1}}{\sum_{i=1}^n T_{AB}} \quad (1)$$

where T_{AB} = trains travelling along section AB and $T_{AB,I=1}$ are trains travelling along section AB when an incident is active.

Similarly, the delay rate is estimated as the proportion of delayed trains out of exposed trains for a given section (Equation (2)).

$$Delayed_{(AB)} = \frac{\sum_{i=1}^n Delayed\ trains_{AB}}{\sum_{i=1}^n Exposed\ trains_{AB}} \quad (2)$$

5.4. Overall delay impact

The impact of an incident is then computed as a combined factor of the rate of occurrence of the incident, the train exposure, the delay rate, and the average delay minutes per incident (Equation (3)). The average delay minutes are computed as the average of the quotient of the total delay minutes and the total delayed trains for all stations and sections in the network. The index, herein referred to as a criticality index computes the expected average delay minutes per train on the network (or section of network) in the event of an incident.

$$CI = \frac{\text{Incident Rate (per day)} * \text{Average Delay minutes (per incident)}}{\text{Total number of trains per day (T)} * \text{Exposure rate} * \text{Delay rate}} \quad (3)$$

This index allows for a comparison of the impact of different incidents. However, important to note is that this index assumes that

incidents and their impact are independent of each other. It also assumes that variation in severity of incidents within the same class e. g., accidents, is negligible.

5.5. Probability of delay

Using R-Studio; a multiple logistic regression model estimating the probability of a train being delayed under different disturbance/incident conditions was developed.

5.5.1. Dependent (Y) and independent variables (X)

The proportion of delayed trains at each location (stretch or station) and day was regressed on the condition of the infrastructure expressed as a binary response (0/1) for each category of incident - where 1 refers to the presence of an active incident and 0 none (Fig. 3). Delay was defined at the train level as the train's deviation from the schedule i.e., minutes late. Notably, the number of passengers and/or the weight of goods carried on these trains, which translates into impact on users, were considered beyond the scope of this study. However, to account for differences in traffic intensity along different sections, the model was weighted by the total number of trains per day per section.

The track type and train/service type were also included as categorical variables of 4 and 2 levels respectively, to control for other factors that influence punctuality. Track type accounts for the network layout including opportunities for trains to pass each other. According to [Sogin et al., \(2013\)](#), there is a linear decrease in delays for every additional sections of double track given a constant volume of traffic. The train (and service) type control for factors like the train lengths (number of carriages) and train running speeds which also influence punctuality ([Harris and Godward 1992](#); [Lai et al., 2010](#); [Landex, 2008](#)).

Tables 2 and 3 show the distribution of data among the aforementioned categories. A passenger train travelling on a single line with no incident was then treated as the baseline/reference scenario.

[Shmueli \(2010\)](#) cites differences in modelling for explanation versus modelling for accurate prediction. Therefore, with a goal to explain the relationship between delays and incidents for the given data, rather than accurately predicting likelihood of delay, the model was trained on the entire sample. This suggests that the results thereof are subject to the quality and structure of the data. The model performance was evaluated using both the pseudo R² test and the rank order test.

5.5.2. Logistic regression model

Equation (1) below shows the general form of the logistic model developed.

$$\log\left(\frac{p(X)}{1-p(X)}\right) = B_0 + B_1X_1 + \dots + B_pX_p \quad (4)$$

where $X = (X_1, \dots, X_p)$ are predictors of p -the probability of delay. In the above model, these include the incident condition, the train/service type, and the track type. Rewriting the above equation; the probability was valued from equation (2) below.

$$p(X) = \frac{e^{B_0 + B_1X_1 + \dots + B_pX_p}}{1 + e^{B_0 + B_1X_1 + \dots + B_pX_p}} \quad (5)$$

Coefficients $B_0, B_1, \dots, B_p$ were estimated using the maximum likelihood method in R.

6. Results and discussion

6.1. Rates of occurrence

Among system failures, signal failures had the highest average occurrence rate of 57 failures per day; nearly double the second most prevalent - switch failures with an occurrence of 27 failures per day. On the extreme end, disruptions from maintenance works had the lowest occurrence rate of approximately 1 incident every 2 days.

Among the 'non-system' incidents, accidents had the highest occurrence rate of 25 accidents per day, followed by natural events at approximately 6 incidents per day while dispatching errors had the least, an average of 1 occurrence every ten days. Column 2 of Table 4 shows incident rates for all incident types.

To account for the influence of weather extremes in Sweden ([Peel et al., 2007](#)), the monthly variation in the daily incident rates were also examined. Snow on track, switch failures and accidents peaked during winter months while signal failures, electrical failures,

Table 2
Distribution of data by track type.

Train/Service type	Frequency	Share (%)
Single track	440,569	29
Double track	296,833	18
Multiple track	16,275	1
Station	503,985	52
Total	1,257,662	100

Table 3
Distribution of data by train/service type.

Train/Service type	Frequency	Share (%)
Freight	387 067	31
Passenger	870 595	69
Total	1 257 662	100

track failures, maintenance works, and natural events peaked during the summer (Fig. 4). As expected, there was no incident of snow on tracks registered for the months of June, July and August for the entire period of study. On the other hand, the largest incident of leaves was experienced in October – an average of 209 incidents per day compared to 17 in September, 34 in November and an average of 3 incidents per day for the rest of the months. The exception was July in which no leaves on track were registered. The rest of the incidents i.e., telecom equipment failures, failures at yards, civil structure failures, and dispatching errors showed a fairly constant rate of occurrence throughout the year.

6.2. Train exposure and delay rate

As expected, train exposure was highest for signal failures with an average of 45% of all trains on the network exposed to a signal failure. This corresponds to the highest incident rate on the network (Table 4). Similarly, dispatching errors with the lowest incident rates resulted in only 0.3% of train exposure i.e., the lowest exposure rate.

In general, the data suggested a high correlation between the rate of occurrence and the incident rate – a logical outcome. The higher the incident rate, the higher the exposure rate. This is illustrated by a superposition of both parameters on the same graph (Fig. 5). It follows therefore that the first step in preventing delays from incidents is to limit their occurrence in the first place. However, slight deviations from this trend can be observed for electrical, telecom equipment, track, yards and civil failures where the rate of exposure is disproportionately higher than the incident rate. Most notable are electrical failures which have the second highest rate of exposure despite a comparatively lower incident rate (four times lower than signals).

This can be attributed to the fact that the functionality of any of these particular assets affects the functionality of other railway subsystems, deeming them ‘critical’. Moreover, data shows that these assets are also associated with averagely higher repair times which may explain the higher exposure rates. Additionally, these assets are also characterized by a large number of components (complexity) and are spread over a larger area (spatial location). For instance, 90% of the signaling system on the Swedish railway network is ATC-2, one of whose main drawbacks is having multiple components including lineside signals, track circuits, balise groups, etc. - where each constitutes a point of failure. Moreover, some components date back to 1980 when the system was first commissioned.

From the seasonal variation in incident rates (Fig. 4), it can be expected that the exposure rates will follow the same pattern.

6.3. Delay impact

Out of the exposed trains, affected trains, for all incident classes, range between 24% and 44% - arguably a minimal variation (Table 4 & Fig. 5). Vehicle failures and accidents had the least delay rate (24%) while dispatching errors (44%) and snow or leaves on track (43%) had the highest. This may be explained by the impact area of these incidents. The former are more localized while the impact area of the latter is large and widespread.

On the other hand, average delay minutes per incident vary greatly among the different incidents with snow or leaves accruing the longest delays i.e., 4.8 hrs (288 min per incident) followed by yards, switches, and signaling respectively. Only vehicle failures lasted

Table 4
Train exposure and delay rates by incident type.

Incident type	Incident rate (per day)	Exposure rate (%)	Delay rate (%)	Average delay minutes/ incident	No. of Trains	Criticality index (Minutes/ Trains)
Signaling	56,9	45,1	29,7	138	1429	41
Switch	26,9	20,9	28,3	156	1429	50
Accident	24,9	12,3	24,3	82	1429	48
Electrical	13,8	31	30,3	133	1429	14
Telecom	11,5	15,3	39,8	108	1429	14
Equipment						
Track	6,3	11,4	28,6	113	1429	15
Natural Events	5,6	3,1	30,7	116	1429	48
Yards	5,4	10,6	34,1	157	1429	16
Snow Or Leaves	0,9	0,2	42,6	288	1429	213
Vehicle	0,8	0,3	24,4	45	1429	34
Civils	0,8	2,1	29,3	69	1429	6
Maintenance works	0,5	0,3	26,2	68	1429	30
Dispatching	0,1	0,3	43,7	106	1429	6

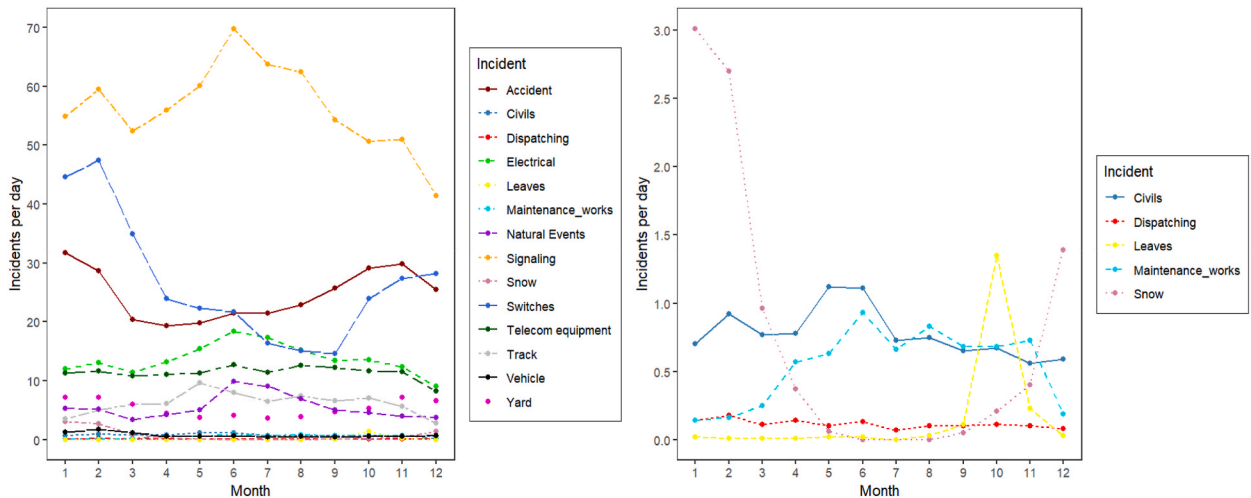


Fig. 4. Monthly variation of incident rate per day.

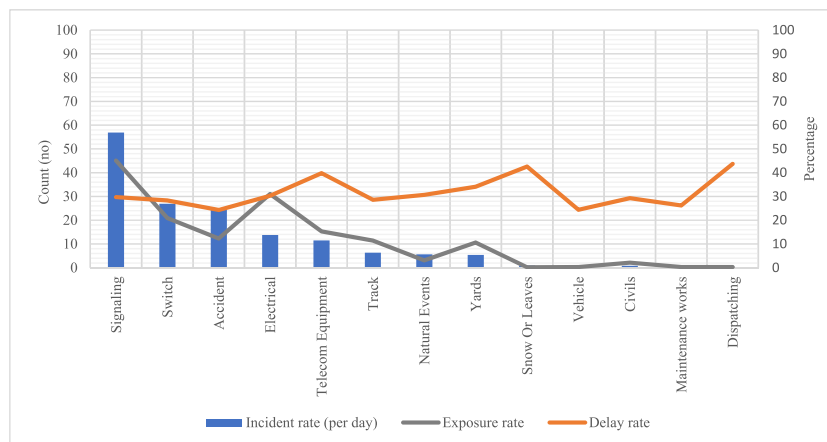


Fig. 5. Variations between exposure, delay and incident rates.

less than 1hr per incident i.e., 0.75 h s. This may be attributed to their respective disruption management protocols. These values differ greatly from those reported by [Nyström and Kumar\(2003\)](#) for a single-line track in the north of Sweden. Values reported in this study are much higher e.g., while this study estimates 2.3 h for signal failures, their study reports only 18 min delay; where we estimate 1.88 h for a track failure, their study reports only 24 min. This may be attributed to the fact that their study only accounts for trains delayed beyond 5 min while we consider all delays. This validates assertions that the use of delay attribution data leads to under-accounting of delays.

However, considering the combined effect as described in Section 5.4, snow or leaves on track, switch failures and accidents were determined as the most critical, resulting in an overall higher impact. This is a slightly different result from the above. Emergent is the fact that using delay minutes only as an indicator of delay impact doesn't provide a full picture as other parameters such as the occurrence and exposure rate are of consequence. This is best illustrated by incidents like dispatching errors, and snow or leaves on track which though having high average delay minutes per incident, rarely occur and their exposure is equally low. Notably, the most critical incidents peak during winter months ([Fig. 4](#)) implying the need to rump up disruption management efforts for these specific failures especially in the winter.

6.3.1. Delay impact viz-a-viz the duration of disruption

Basing on a randomly selected stretch along the network, a comparison of the evolution of train delays over the disruption period was made for different incidents. [Fig. 6](#) shows that for both incidents all (100%) the passing trains are delayed, however, different patterns emerge for average delay minutes per train per day for signal and track failures. In signal failures, they increase from the onset while they decrease for track failures. These differences may be attributed to differences in disruption management protocols and the system's inherent ability to react and respond to the two types of failures/incidents.

This possibly highlights the need to incorporate the effect of delay management protocols in such a framework. However, in the absence of related data, this is considered beyond the scope of this study, and the average delay minutes for all trains over the disruption length was used instead (refer to section 5.4 and section 6.1).

6.4. Risk of delay

While the exposure rate, delay rate and average delay minutes per incident evaluate the impact of incidents at the network level, they do not depict the impact on individual trains. Therefore, this was modelled as a probability of delay.

Table 5 shows model parameters for estimating this probability of delay under different incident scenarios. A passenger train travelling on a single track was treated as the baseline/reference scenario. The logistic model showed an ‘average’ performance with a McFadden’s pseudo R^2 value of 0.102 and a KS-statistic of 10% achieved within the 3rd decile, capturing up to 40% of data. According to the model, the influence of all incident classes including service/train type and track layout are statistically significant.

Based on model parameters and equation (2) above, the inherent risk of delay of a passenger train travelling on a single track with no incidents was estimated at 28%. This underscores the fact that the punctuality of trains is influenced by other factors besides incidents on the railway - which is in line with several other studies (Olsson and Haugland, 2004; Xia et al., 2013). A similar train travelling on double track showed nearly 50% (half) reduction in odds of delay, which is in stark contrast to multiple tracks with approximately 90% increased odds of delay while, the odds of delay at a station are only 6% less than they are for a train travelling along single track. The unexpected increase in odds of delay for stations compared to double track may be attributed to the increased complexity of the track layout at stations, hence making them highly susceptible to failure. Additionally, the increased interaction of trains at these sections makes them hotspots for secondary delays.

Interestingly, a freight train travelling on a single line had close to 72% reduced odds of delay. The much lower probability of delay (22%) associated with freight trains, can be explained by the fact that the schedule for freight trains is more flexible in Sweden in that freight trains maybe allowed to run early and or be cancelled (Palmqvist et al., 2022). Arguably, this gives the dispatcher the option to avoid major incidents, which is not the case with passenger trains that must endeavor to stick to the timetable in order to meet passenger expectations.

According to the model, the odds of delay are increased for all incidents except maintenance works, switch failures, dispatching errors, and accidents for which the model showed (minimal) reduced odds of delay. Besides accidents which constitute 16% of the data, the rest constitute comparatively low proportions in the sample – which may explain the result. However, another explanation could be the disruption management practices applied for these incidents which may include cancellation of trains, temporary speed restrictions, rerouting of trains and/or drivers speeding up to compensate for a delay, among others. Due to limitations in data, as explained in Section 6.3.1, these factors were not incorporated, however future studies could examine their impact on delay propagation. Snow or leaves and telecom equipment failures showed a substantially higher impact i.e., 91% and 77% increased chance of train delay respectively, compared to a range of 1–20%, at a 95% confidence interval, increased odds of delay for other incidents. The model also showed that trains experience the same odds of delay under electrical and signal failures and similarly, the odds of delay under natural events are the same as those during civil infrastructure failures. This is logical given that civil structures such as embankments are most adversely affected by natural events such as floods. In general, the results also speak to the impact area of the different incidents. Incidents whose impact spreads over a larger area of the network such as snow or leaves on track, telecommunication equipment, signal failure, and electrical failures show higher odds of delay than those whose impact is more localized.

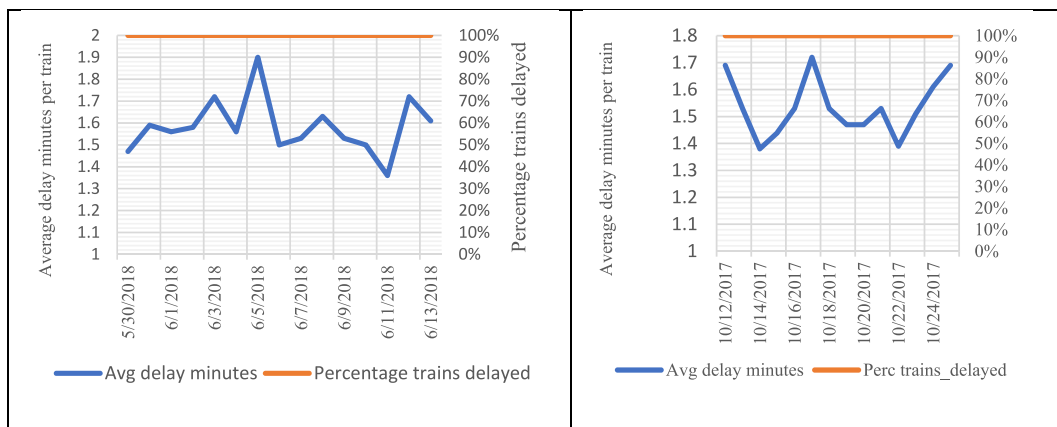


Fig. 6. Comparison of the evolution of train delays over the disruption length (14 days) for a signal failure (L) and track failure (R) at Pärp-Ättekulla (L).

Table 5

Parameter estimates for the logistic regression model.

X-variables	Estimate	Std. Error	z value	Pr ($> z $)	SS	Percentage decrease/ increase in odds	Lower bound (95% cf)	Upper bound (95% cf)
(Intercept)	−9,49E-01	9,52E-04	−996,899	<2e-16	***			
Infrastructure/ System								
Snow or leaves	6,45E-01	5,95E-03	108,458	<2e-16	***	90,6	88%	93%
Telecom equipment	5,69E-01	7,74E-04	735,247	<2e-16	***	76,7	76%	77%
Yard	1,78E-01	9,45E-04	188,081	<2e-16	***	19,5	19%	20%
Signals	1,26E-01	5,98E-04	210,019	<2e-16	***	13,4	13%	14%
Electrical	1,24E-01	6,41E-04	193,72	<2e-16	***	13,2	13%	13%
Civil	9,30E-02	2,05E-03	45,269	<2e-16	***	9,7	9%	10%
Track failure	5,86E-02	9,38E-04	62,464	<2e-16	***	6	6%	6%
Vehicle	2,58E-02	5,77E-03	4471	7,77E-06	***	2,6	1%	4%
Maintenance	−1,79E-02	5,17E-03	−3472	0,000516	***	−1,8	−3%	−1%
Switches	−9,23E-02	7,45E-04	−123,99	<2e-16	***	−8,8	−9%	−9%
Other								
Natural Events	9,00E-02	1,68E-03	53,474	<2e-16	***	9,4	9%	10%
Dispatching	−4,88E-02	5,11E-03	−9547	<2e-16	***	−4,8	−6%	−4%
Accidents	−1,50E-01	9,42E-04	−158,948	<2e-16	***	−13,9	−14%	−14%
Track type								
Multiple track	6,34E-01	1,58E-03	401,655	<2e-16	***	88,5	88%	89%
Station	−6,43E-02	8,93E-04	−71,977	<2e-16	***	−6,2	−6%	−6%
Double track	−6,29E-01	1,25E-03	−502,761	<2e-16	***	−46,7	−47%	−47%
Train/Service type								
Freight	−1,26E+00	9,28E-04	−1357,28	<2e-16	***	−71,6	−72%	−72%

7. Conclusion

In this study, we demonstrate how historical train operation data, specifically train run data and failure rates can be used to examine the influence of railway incidents on train delays both at the network level and for individual trains. At the network level, the study proposes two measures to evaluate the sensitivity of trains to different incidents-the exposure and delay rate. While the former evaluates the reach of an incident i.e., the proportion of trains that travel over a section with an active incident, the latter is specific to punctuality-evaluating the proportion of trains that are affected by the incident. Exposure to incidents cannot be dissociated from their occurrence, while the duration of delay is crucial for understanding the impact of an incident. Therefore, the study proposes a comprehensive measure combining all four parameters (i.e., rate of occurrence/incident rate, exposure rate, delay rate and average delay minutes) to evaluate and compare the criticality of incidents to train delays. For the case study, snow or leaves on track were, by far, the most critical resulting in the longest normalized delay minutes at the network level.

For individual trains, the study developed a model to explain the risk of delay on all trains travelling on the network in the event of a given failure. It was established that snow or leaves still posed the highest risk of delay to individual trains and by the ranking of risk, it can be inferred that the risk is driven by the area of impact of the different incidents.

The main limitation of this study, which also constitutes the direction for further research is that the severity of incidents and the effect of simultaneously occurring incidents are not accounted for. The proposed framework assumes that all incidents occur independently of each other. Moreover, in mapping incidents to trains, the impact area is not explicitly accounted for, in the raw data a delay is only mapped to an incident within the same section. This may result in an underestimation of the delay impact. In future research, impact areas will be considered. Other factors influencing train delays could be quantified and incorporated in the model to make it more explainable. Specifically, results need to be tested in consideration of capacity utilization levels in order to determine the extent to which the low levels of risk determined will hold, under different capacity utilization levels. According to [Olsson and](#)

Haugland (2004), above a given capacity utilization threshold, the impact of incidents on train punctuality becomes more eminent. The disruption management practices employed under different scenarios and the system's ability to react and restore the initial operating conditions also need to be accounted for in estimating the impact of incidents on train delays as demonstrated by differences in the average delay minutes per train over the duration of the incident. Relatedly, no consideration is made of the exact point along an incident's resilience curve that the train runs and its impact on delay. Arguably, the delay during the degradation phase is likely to be bigger than the delay during the response or recovery phase. Accounting for this will improve the results of such a study.

The main contribution of this paper is a method of assessing the impact of railway incidents on train punctuality that is based on historical operation data i.e., failure reports and train run data. Subject to availability of data, these methods may be applied to investigate single or multiple types of incidents on any railway network. Moreover, the model to estimate probability of delay can be extended to include other determinant factors for punctuality.

Understanding the sensitivity of train operations to incidents on the network points towards 'weak points' within the railway system and so provides insight into areas of focus in improving punctuality of the railway system. The risk of delay from incidents may also provide insight into key operations planning processes such as the simulation process and where to insert delay distributions. It also provides insight into prioritization of recovery actions during delay management and may provide guidance on the allocation of buffer times during initial timetable development in that one may choose to include larger buffers at locations with a component whose failure impacts delays more. Stenström et al. (2016) also proposed a quick method for estimating the prevented train delays from infrastructure inspections which requires the probability of a failure resulting in train delays as an input parameter.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

Funding for this research and the raw data used was provided by the Swedish Transport Administration under grant numbers TRV 2022/107738, TRV 2023/9635, and TRV 2024/2731. The authors acknowledge the contribution of the Railway research group at LTH for their constructive review and comments.

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